

BUSINESS INSIGHTS AND RECOMMENDATIONS REPORT

1. Executive summary

This analysis examined a loan application dataset to identify the key factors associated with loan approval and to prepare a reliable dataset for predictive modelling. The business problem is to better understand how applicant and application characteristics influence loan approval, with the aim of supporting more consistent, objective, and evidence-based lending decisions. The main objective was therefore to conduct advanced exploratory and statistical analyses, identify meaningful features and relationships, address data quality issues, and prepare a machine-learning ready dataset for week 4. The dataset contains 614 loan applications, of which 422 were approved and 192 were rejected. This represents an approval rate of 68.73% and a rejection rate of 31.27%, respectively. The conducted statistical analysis identified significant relationships between loan status and several applicant characteristics, including married status, education, and property area. CreditRiskCategory also showed a strong association with loan status. However, because this variable was derived from Credit_History, it was removed from the final modelling dataset to avoid introducing redundant information.

Correlation analysis provided further insight into the relationships between the financial variables. ApplicantIncome and TotalIncome showed a strong positive correlation of 0.8930, indicating that the two variables contain very similar information. LoanAmount was moderately correlated with both ApplicantIncome (0.5709) and TotalIncome (0.6246). In contrast, TotalIncome and DebtToIncomeRatio showed a moderate negative correlation of -0.3882. In addition, several data quality issues were also identified during the analysis. Credit_History had the highest number of missing observations, with 50 missing values (8.14%), followed by Self_Employed with 32 missing values (5.21%). LoanAmount and DebtToIncomeRatio each had 22 missing observations (3.58%). These missing values were addressed during the dataset refinement process. Loan_ID was removed because it serves only as an identifier and does not provide meaningful predictive information. Categorical variables were encoded into numerical form, while numerical predictors were scaled to ensure that the features were on comparable scales for modelling.

After preprocessing and validation, the final modelling dataset contains 614 observations and 23 variables, comprising 22 predictor variables and the Loan_Status target variable. The validation process confirmed that there are no remaining missing values. Loan_Status was encoded as 1 for approved applications and 0 for rejected applications. Overall, the dataset has been successfully cleaned, refined, and prepared for predictive modelling. The identified relationships provide useful insights into the factors associated with loan approval, while the preprocessing steps ensure that the data is suitable for machine-learning algorithms. The refined dataset is therefore ready for the week 4 predictive modelling stage.

2. Key analytical findings

2.1 Loan approval distribution

There were 422 approved applications (68.73%) and 192 rejected applications (31.27%). Approved applications therefore form the majority of the dataset. This class distribution should be considered during week 4 because accuracy alone may not provide a complete picture of model performance.

2.2 Categorical associations with loan status

The Chi-square test showed statistically significant associations between Loan_Status and Married status ($p = 0.034394$), Education ($p = 0.043100$), and Property_Area ($p = 0.002136$). These findings suggest that household characteristics, educational status, and property location contain information that may be useful when predicting loan outcomes in this dataset. Other variables did not show statistically significant associations at the 5% level. Gender had $p = 0.708653$, Dependents $p = 0.367851$, Self_Employed $p = 1.000000$, IncomeBand $p = 0.889746$, and FamilySizeGroup $p = 0.784945$. A non-significant result does not prove that a variable has no predictive value, it indicates that the test did not provide sufficient evidence of an association in this sample.

2.3 Credit history and feature evaluation

CreditRiskCategory showed a strong statistical association with Loan_Status ($X^2 = 180.0665$, $p < 0.001$). However, the feature was derived from Credit_History. Keeping both would duplicate information and make the model less interpretable. CreditRiskCategory was therefore removed, while Credit_History was retained.

2.4 Numerical relationships

ApplicantIncome and TotalIncome had a strong positive correlation, with correlation coefficient of $r = 0.8930$. This is expected because TotalIncome incorporates applicant and coapplicant income. LoanAmount was positively correlated with ApplicantIncome ($r = 0.5709$) and TotalIncome ($r = 0.6246$). TotalIncome and DebtToIncomeRatio had a moderate negative correlation ($r = -0.3882$). These relationships should be considered when interpreting model coefficients and feature importance.

2.5 Mann–Whitney U Test

The test produced a statistic of 38,267.0 and a p-value of 0.576442. Since the p-value is greater than 0.05, the analysis did not provide sufficient evidence of a statistically significant difference between the two groups examined.

2.6 Feature refinement

The refinement stage retained meaningful engineered features including TotalIncome, IncomeBand, DebtToIncomeRatio, and FamilySizeGroup. Loan_ID was removed because it is an identifier, and CreditRiskCategory was removed because it duplicated Credit_History. Categorical variables were one-hot encoded, numerical variables were standardised, and Loan_Status was encoded as a binary target.

3. Business implications

The findings suggest that a data driven lending process should consider a combination of credit history, applicant characteristics, financial information, and application details rather than relying on a single factor when assessing loan applications. Credit_History appears to be particularly significant and should therefore receive careful attention during predictive modelling. The analysis also identified a significant relationship between Property_Area and Loan_Status, indicating that loan outcomes differ across property areas within the available dataset. However, this finding does not mean that location itself causes loan approval or rejection. Further analysis would be required to understand the factors that may contribute to this pattern. The strong correlation between ApplicantIncome and TotalIncome indicates that these variables contain overlapping financial information. Although they represent different business concepts, their relationship should be considered during model development to reduce the potential impact of multicollinearity and to maintain model interpretability. The missing values identified in Credit_History are also important from a business perspective. Since credit history can play a significant role in lending decisions, incomplete information in this field may affect the reliability of automated or semi-automated decision support systems. Improving the collection, verification, and completeness of important applicant information could therefore strengthen data quality and support more reliable lending decisions in the future. Overall, the findings highlight the value of combining multiple sources of applicant and application information when assessing loan applications. These insights provide a useful foundation for the predictive modelling stage and can help inform the development of more consistent and evidence-based lending processes.

4. Recommendations

- Prioritize Credit_History as a candidate predictor in week 4 modelling and carefully monitor the effect of missing credit history information,
- Include Property_Area, Education, and Married status as candidate predictors because they showed statistically significant associations with Loan_Status,
- Retain meaningful financial variables such as ApplicantIncome, TotalIncome, LoanAmount, and DebtToIncomeRatio while monitoring their correlations,
- Keep Loan_ID excluded from predictive modelling because it is an identifier rather than a meaningful applicant feature,
- Do not use CreditRiskCategory together with Credit_History because the engineered category is derived from the same underlying information,
- Do not automatically remove variables solely because their individual statistical tests were non-significant, rather evaluate their contribution during model development,
- Use accuracy together with precision, recall, F1-score, confusion matrix analysis, and Receiver operating characteristic – area under the curve (ROC-AUC) where appropriate during week 4,
- Maintain a reproducible record of imputation, encoding, scaling, and feature refinement decisions.

5. Dataset limitations

- The dataset contains 614 observations, which is relatively small for developing a highly generalizable predictive system,

- Many variables contained missing values. Credit_History had 50 missing values, Self_Employed had 32, and LoanAmount and DebtToIncomeRatio each had 22,
- Imputation addresses the missingness computationally but introduces assumptions,
- The target distribution is not perfectly balanced, with 68.73% approvals and 31.27% rejections and this may influence model performance,
- The dataset may not represent all applicants, lenders, geographic areas, or economic conditions and the findings should therefore not automatically be generalized to other populations,
- The available variables do not capture every factor that may influence real world lending decisions, such as detailed repayment history, credit-score components, employment stability, or institution specific risk rules,
- Statistical associations do not establish causation. The findings describe patterns within the available data rather than proving that a particular characteristic causes approval or rejection.

6. Next steps

The refined dataset will provide the foundation for week 4 predictive modelling stage. The next step will be to separate the predictor variables from the Loan_Status target and divide the data into appropriate training and testing sets. Where feature scaling is required, the scaling parameters should be learned from the training data and then applied to the test data. This approach helps prevent data leakage and ensures that the model is evaluated fairly on unseen data. Several classification algorithms can then be developed and compared, including Logistic Regression, Decision Tree, and Random Forest models, along with other suitable classification techniques. The models should be evaluated using a range of performance measures, including accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC where appropriate. Using multiple metrics will provide a more balanced assessment of model performance rather than relying on accuracy alone.

Feature importance and model interpretation should also be examined to identify the variables that contribute most strongly to loan approval predictions. This will help connect the modelling results to the business findings identified during the exploratory and statistical analysis. Finally, the selected model should be evaluated not only on its predictive performance but also on its stability, interpretability, and suitability for the lending context. A model that performs well but is difficult to interpret or unreliable on new data may have limited practical value. The goal should therefore be to identify a model that provides a good balance between predictive accuracy, reliability, transparency, and business usefulness.

7. Conclusion

The week 3 analysis provided a clearer understanding of the loan prediction dataset and established a strong, evidence-based foundation for the predictive modelling stage. The statistical analysis identified significant relationships between loan status and selected categorical variables, while the correlation analysis highlighted important relationships among the financial features. The data refinement process also addressed key quality and preprocessing requirements. Missing values were handled, the non-informative Loan_ID variable was removed, and the derived CreditRiskCategory variable was excluded to avoid redundant information.

Categorical variables were encoded into numerical form, numerical predictors were scaled, and the Loan_Status target was converted into a suitable format for machine-learning. Following these steps, the final modelling dataset contains 614 observations and 23 variables, consisting of 22 predictors and one target variable. Final validation confirmed that there are no remaining missing values in the dataset. Overall, the dataset is now clean, structured, and ready for week 4 predictive modelling. The next stage will involve training and evaluating classification models to determine how effectively the identified applicant and application characteristics can be used to predict loan approval outcomes.